

DUSTBADGE · DESIGN DECISIONS REGISTER

DustBadge design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Worn version carried into the design: foam pad on the BOM (USD 91.50, USD 0.50 over the target); value engineering section restated; target change proposed
0.4	2026-10-02	Amish Chadha	Amish approved the recommendations for all ten open decisions (2026-10-02); DBG-DDR-003 accepted; moved to decisions made
0.3	2026-10-01	Amish Chadha	Dr. Geeti Chadha added as a contributor and the project moved to the BioMedical (healthcare) area (decided by Amish)
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

DustBadge design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision	Options	Recommendation
1	Value-engineering target. The foam pad of the worn version (USD 0.50, BOM line 13) takes the estimated cost to USD 91.50, USD 0.50 over the USD 91 target, so R15 is not met. <code>budget_usd</code> is unchanged at 91.	(a) raise the target to USD 92; (b) keep USD 91 and look for USD 0.50 of savings when parts are bought; (c) keep USD 91 and accept R15 as not met.	(a), so that the target covers the worn version; the optional lanyard (about USD 1) would still sit outside it. Proposed, awaiting Amish.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The cell, including its protection board, is no larger than 50 x 34 x 11.5 mm, its lead leaves at one short side, and its plug polarity matches the charger	The cell ribs, the swelling allowance (0.7 mm in the bench version, 1.7 mm with the foam pad of the worn version) and the board notch are sized for it	DBG-DDR-003, P8
2	The charger module is no larger than 11 x 19 mm with its USB-C on a short edge, and its temperature window is 0 to 45 °C	It must fit beside the sensor with its USB-C in the opening, and the charge stop is part of the safety case	DBG-DDR-003, P6
3	The boost module (11.4 x 9 mm or less, with an enable input) and the humidity breakout (10 x 10 mm or less, sensor at one edge)	They are taped into a strip 17 mm wide beside the sensor; the humidity sensor must sit over its vent	DBG-DDR-003, P6
4	Where the particle sensor's cable leaves it, and the cable length	The model leaves room above and beside the sensor between the stop ribs; a connector on another face would need the ribs moved	DBG-DDR-003, P5
5	The spring clip's base leaf is flat and wide enough for two holes 12 mm apart	The clip is held by two M2 x 8 screws	DBG-DDR-003, P9
6	The pilot hole for the M2 thread-forming screws in PETG (1.7 mm assumed), from the screw maker's data	Too large a hole strips; too small cracks the boss	DBG-DDR-003, P1

■ Value engineering

Value-engineering target: USD 91 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 91.50 (USD 0.50 over the target). The estimate rests on indicative prices.

Main cost drivers: the optical particle sensor (USD 48.00, 52 % of the cost), the 2,000 mAh LiPo cell (USD 13.00; it added USD 3), the controller and BLE module (USD 10.00) and the carrier board (USD 9.00).

Savings worth trying: confirming prices when parts are bought. A 1,500 mAh cell would save USD 3 but returns R7 (a full shift) to at risk. The optional breakaway lanyard (decided on 2026-10-02 for TRL 4) would take the estimate about USD 1 over the target, and the foam pad of the deeper worn version (USD 0.50) is already in the estimate and is what takes it over the target.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D7: SPS30-class sensor, silica shown only as a labeled estimate, US OSHA and MSHA default limits, continuous sampling, worker-owned data, surface quarries and stone fabrication first, downward screened inlet with a yellow shell and clip	Amish: "i accept all your recommendations, go with them across all repos."	DBG-DDR-001, DBG-DDR-002
2026-09-25	D8 to D11: 2,000 mAh cell; R3 reference rule (4.2 L/min cyclone over two shifts at low-dust, high-silica sites); sun use rule; R2 restated to 1 mg/m ³ with an over-range flag	Amish, same instruction	DBG-DDR-002
2026-09-26	Value-engineering target set to \$91 to match the priced BOM (O3)	Amish: "i approve all the budget items."	DBG-DDR-002 v0.2
2026-10-01	Dr. Geeti Chadha added as a contributor (CONTRIBUTORS.md, README Credits) and the project moved to the BioMedical (healthcare) area, with soft, non-clinical wording: a research and educational prototype, not a medical device	Amish: "yes add Dr. Geeti Chadha to breathebox and dustbadge and make those both healthcare projects"	project.yaml , CONTRIBUTORS.md , README.md , DBG-PRB-001 v0.7, DBG-PRC-001 v0.8
2026-10-02	Design for construction accepted: P1 to P10 as made; the TRL 4 drop test also checks that the boost module, humidity board and motor stay on their foam tape	Amish: "i approve your recommendations for all 555 open decisions."	DBG-DDR-003, Table 1
2026-10-02	Cell swelling: the 0.7 mm allowance (a) is accepted only for bench work with nobody wearing the badge; before anyone wears it, (b): the badge is deepened 1 mm with a foam pad behind the cell and R9 is relaxed to 122 g. Return to (a) only if the bought cell's datasheet states swelling under about 6 %	Amish: "i approve your recommendations for all 555 open decisions."	DBG-DDR-003, A1
2026-10-02	On and off: firmware only (the boost switched off through its enable input, the controller asleep, waking on USB power), with a firmware low-voltage cutoff set above the cell protection threshold so the cell is never run down to the protection cut-out	Amish: "i approve your recommendations for all 555 open decisions."	DBG-DDR-003, A2
2026-10-02	R9 mass margin: accepted, the prototype is weighed at TRL 4; with the deeper shell of the swelling decision the R9 limit is 122 g	Amish: "i approve your recommendations for all 555 open decisions."	DBG-DDR-003, A3; DBG-CAL-001 [J2]
2026-10-02	First co-design partner: a university industrial hygiene group that already samples respirable crystalline silica, for example one linked to a NIOSH Education and Research Center, working with a stone fabrication shop or surface quarry as the site. This is the first candidate type to approach, not an agreed partner	Amish: "i approve your recommendations for all 555 open decisions."	DBG-DDR-001, O1

Date	Decision	Decided by	Record
2026-10-02	Site silica fraction: by default two or three cyclone filter samples per site go to an accredited laboratory (X-ray diffraction, as in NIOSH Method 7500), and no silica estimate is shown until a site fraction exists; field infrared analysis is considered later where a partner has the instrument	Amish: "i approve your recommendations for all 555 open decisions."	DBG-DDR-001, O2
2026-10-02	Clear front window: renders only; decided at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 1
2026-10-02	Sensor fan in the renders: kept, with a caption saying it is illustrative	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 2
2026-10-02	Side grip ribs: appearance only; if adopted at TRL 4 they are recessed so the badge stays 64 mm wide	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 4
2026-10-02	Lanyard: an optional lanyard line is added at TRL 4, specified as breakaway only	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 5